

Exp No:28

COUNTING SEMAPHORE IN SYSTEMC

PROGRAM:

```
#include <systemc.h>

SC_MODULE(counting_semaphore) {

    int sem;

    void task() {
        while (true) {
            wait(1, SC_SEC);

            if (sem > 0) {
                sem--;

                cout << "Task Accessing Resource at "
                     << sc_time_stamp()
                     << " Remaining: " << sem << endl;

                wait(2, SC_SEC);

                sem++;
            }
        }
    }

    SC_CTOR(counting_semaphore) {

        sem = 2;

        SC_THREAD(task);
        SC_THREAD(task);
    }
}
```

```

        SC_THREAD(task);
    }
};

int sc_main(int argc, char* argv[]) {

    counting_semaphore obj("Counting");

    sc_start(10, SC_SEC);

    return 0;
}

```

OUTPUT

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL <https://www.edaplayground.com/x/5Vg4>. The interface includes a sidebar with navigation options like 'Languages & Libraries', 'Tools & Simulators', and 'Examples'. The main area is divided into two code editors: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' editor contains the following code:

```

23 }
24
25 SC_CTOR(counting_semaphore) {
26
27     sem = 2;
28
29     SC_THREAD(task);
30     SC_THREAD(task);
31     SC_THREAD(task);
32 }
33 };
34
35 int sc_main(int argc, char* argv[]) {
36
37     counting_semaphore obj("Counting");
38
39     sc_start(10, SC_SEC);
40
41     return 0;
42 }

```

The 'design.cpp' editor contains the following code:

```

1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>

```

Below the code editors is a 'Log' section showing the execution output. The output includes the following text:

```

[2026-08-25 04:06:43 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

Warning: (w505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: ../../src/sysc/kernel/sc_object_manager.cpp:153

Warning: (w505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: ../../src/sysc/kernel/sc_object_manager.cpp:153

Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0

```

The bottom of the screenshot shows a Windows taskbar with various application icons and a system clock indicating 09:33 on 25-08-2026.